

Ch. 6 — P02 (K) Random Number Streams and Reproducibility

Problem Bank | Chapter 6 | Type: K | Difficulty:

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Random number management is critical for reproducible and comparable simulation experiments.

Tasks

1. **Stream independence:** Using `np.random.default_rng(42).spawn(3)`, generate 3 child generators. Draw 10 `exponential(mean=1)` samples from each and verify (by inspection) that they produce different sequences despite sharing a parent seed.
2. **Reproducibility:** Show that running the same SimPy M/M/1 model with the same seed twice produces identical results. Then show that changing the seed by 1 produces different (but still valid) results.
3. **Stream synchronisation:** You are comparing two service policies: FCFS and LCFS. Both should use the same sequence of interarrival times (same randomness in arrivals) but may use different service time sequences. Write code that:
 - Creates one `arr_rng` shared between both policies
 - Creates two separate `svc_rng_fcfs` and `svc_rng_lcfs`
 - Simulates 200 customers under each policy
 - Reports mean sojourn time $\bar{W}$ for each

Do you expect $\bar{W}$ to differ between FCFS and LCFS? Why or why not? (*Hint: for M/M/1, Little's Law implies $\bar{W}$ is the same regardless of discipline as long as the server is non-preemptive.*)

4. **Anti-pattern:** A student writes the following code:

What is wrong with this? What will `results` contain? Rewrite the loop correctly.